Using Fossil Fuels

Chapter 3 Unit 2 Guided Notes · Grade 4 Using Natural Resources for Energy

Name: _____ **Date:** _____

BIG QUESTION **What are some costs and benefits of using fossil fuels?**

KEY WORDS

extract — to take a resource _____ the ground.

refine — to change crude oil into _____ useful products.

combustion — _____ a fuel (burning).

FROM GROUND TO USE

Number the steps in order from 1 to 4.

STEP	ORDER
extract (take it from the ground)	_____
transport (move it)	_____
refine (change it into useful products)	_____
burn it to release energy	_____

CHECK YOUR THINKING

Why is the order of these steps important?

Electricity Pathway

Name: _____ **Date:** _____

ELECTRICITY PATHWAY

Burning fuel heats water into steam. The steam spins a turbine and turns a generator.

BURN THE FUEL

↓

HEAT WATER INTO STEAM

↓

SPIN A TURBINE

↓

TURN A GENERATOR

↓

ELECTRICITY

DRAW / LABEL

Draw the pathway from burning fuel to electricity.

draw the electricity pathway here

Benefits, Limitations, Stewardship

Name: _____ **Date:** _____

BENEFITS AND LIMITATIONS

BENEFIT	LIMITATION
heating homes and cooking	_____
powering engines (cars, trucks)	_____
making electricity	_____
mining can be dangerous	_____

CAREFUL USE

Extraction and use can affect land, water, and air.

An oil spill can harm _____ and wildlife.

Burning fossil fuels releases _____ and other substances into the air.

Stewardship means taking care of resources and the places we live.

CHECK YOUR THINKING

Why might someone choose a renewable resource instead of a fossil fuel?

CONNECT *Chapter 3 connection: fossil fuels give us energy and products, but they have real costs to consider.*